

MUHAMMAD MAHEEM

AI Developer & Machine Learning Engineer

mirza.muhammad.maheem@gmail.com +92 300-8714141 LinkedIn GitHub Portfolio Lahore, Pakistan

ABOUT ME

21-year-old AI Developer and Machine Learning Engineer focused on building intelligent systems that solve real-world problems. Experienced across the entire development lifecycle—from data preprocessing and model training to deploying production-ready applications with robust authentication and responsive UIs. Specialized in RAG systems, Computer Vision pipelines, and Reinforcement Learning.

PROFESSIONAL EXPERIENCE

Frontend Developer

Artificial Automation Business Solutions (AABS)

Sep 2025 – Nov 2025 · Part-time · Remote

- Developed responsive UI components for a full-stack SaaS Management Dashboard using Next.js 15, React 19, and TypeScript.
- Implemented secure user authentication flows with NextAuth, analytics visualizations, and subscription management interfaces.
- Collaborated asynchronously with backend teams to integrate RESTful APIs and ensure seamless data flow in production environments.

FLAGSHIP PROJECTS

RAG-Based PDF Chatbot (Production-Ready)

- Developed a secure end-to-end RAG system using FastAPI, Next.js 14+, and Groq (Llama 3 70B).
- Built a robust document pipeline: text chunking → Sentence Transformers embeddings → ChromaDB vector storage.
- Implemented JWT authentication, isolated user workspaces, and efficient PDF retrieval for context-aware responses.

Cultural Heritage AI: Hadith Chatbot

- Indexed 39,038 authentic Hadiths using FAISS vector search and LangChain for semantic retrieval.
- Developed a Flask-based backend capable of answering complex queries based on religious texts with high accuracy.
- Optimized for fast inference and precise cross-referencing across large-scale textual datasets.

Face Recognition Attendance with Anti-Spoofing

- Built a liveness-aware recognition system using Dlib/OpenCV with blink detection to prevent photo-based spoofing.
- Designed an enrollment system that captures facial expressions to enhance validation security.
- Created a Streamlit GUI for real-time validation, user management, and automated attendance logging.

Tic-Tac-Toe AI via Deep Q-Network (DQN)

- Trained a Reinforcement Learning agent in PyTorch over 100,000 self-play episodes.
- Implemented experience replay and epsilon-greedy exploration to achieve a near-optimal game strategy (win/draw rate >98%).
- Developed model persistence scripts for seamless loading and real-time interaction via a CLI interface.

Additional Technical Projects

Value Notes: Developed a Chrome extension with React 18, TypeScript, and Tailwind CSS, featuring real-time Firebase synchronization.

NLP Personality Prediction: Built an SVM-based model to classify introversion/extroversion traits with semantic feature extraction.

Bank Management System: Designed a functional banking application using OOP principles in Python and Streamlit.

TECHNICAL SKILLS

AI & Machine Learning

Python • PyTorch • Scikit-learn • Computer Vision (OpenCV, Dlib) • RAG Systems • LangChain • ChromaDB • FAISS • Reinforcement Learning

Web Development

Next.js 14/15 • React 18/19 • FastAPI • TypeScript • Tailwind CSS • Firebase • Streamlit • REST APIs

Tools & Soft Skills

Git/GitHub • OOP • Linux • Data Structures • Problem-Solving • Technical Communication • Self-Directed Learning

Interests: Astronomy, Sketching, Gaming, and Search Engine Optimization (SEO).

EDUCATION

Superior University

2024 – 2028

BSc in Artificial Intelligence (4th Sem, CGPA: 3.58/4.0)

Lahore, Pakistan

Focus: Machine Learning, Deep Learning, NLP, Computer Vision, DSA, Linear Algebra, Probability & Statistics.